ADM Unit 4 | Mission Brief 4.5

Smart Factory Integration Capstone

Propose and demonstrate a secure, scheduled, and resilient smart-factory system for an aerospace product.

Customer	Department	Difficulty	Time
Lockwood Aerospace Manufacturing	Smart Factory Program	Advanced	3-5 class periods

Production Goal: Design and defend an integrated aerospace manufacturing system.

Background

Lockwood Aerospace Manufacturing is selecting a team to design its next small smart factory. The proposal must integrate automation, data, quality, scheduling, packaging, cybersecurity, and contingency planning into one coherent system.

Learning Objectives

- Integrate technical and operational manufacturing systems
- Create production and information-flow models
- Use trade studies to justify decisions
- Present and defend a complete system proposal

Equipment

- ADM workcell and/or digital twin
- Layout and process-mapping tools
- Demand and capacity data
- Cybersecurity and risk templates
- Presentation software
- Engineering notebook

Student Instructions

1. Select one aerospace product or component from the provided options.
2. Define product specifications, demand, due dates, and quality checks.
3. Create a factory layout showing material flow, robot/conveyor stations, inspection, packaging, and shipping.
4. Create an information-flow diagram showing sensors, production data, operator inputs, and protected systems.
5. Develop a production schedule and calculate expected throughput.
6. Program or simulate one critical automated sequence.
7. Create a cybersecurity risk assessment with at least five controls.
8. Create a resilience plan for one machine failure and one supplier delay.
9. Build a decision matrix comparing your design with one alternate concept.
10. Present the proposal and answer questions from a review panel.

Engineering Requirements

- Meet the stated demand target or document the capacity gap
- Include automation, inspection, packaging, scheduling, cybersecurity, and resilience
- Provide at least one tested or simulated automation sequence

- Support major decisions with data or a trade study
- Include clear safety and reset procedures

Constraints & Safety

- Use available classroom tools and realistic capacity assumptions
- No safety or quality requirement may be removed to improve throughput
- Protect sensitive production data in the system design
- Presentation time is limited to eight minutes

Deliverables

- Factory layout
- Material and information flow diagrams
- Production schedule and throughput calculation
- Automation code or simulation evidence
- Cybersecurity risk plan
- Resilience plan
- Decision matrix
- Eight-minute presentation
- Reflection

Reflection Questions

1. Which subsystem most limited factory performance?
2. What tradeoff had the greatest effect on your design?
3. How did cybersecurity influence physical layout or workflow?
4. What would you validate before purchasing equipment?

Engineering Support

Engineering Tip
A strong smart-factory design connects physical flow, information flow, decision logic, and recovery planning.

Common Mistake
Adding more technology without defining interfaces and ownership can make a factory less reliable, not smarter.

Stretch Goal
Create a cost estimate and calculate the expected payback period for the proposed automation.

Career Connection
Smart manufacturing systems engineers integrate robotics, software, data, quality, cybersecurity, and operations.

Before You Submit

- Notebook evidence included
- Requirements checked
- Testing documented
- Reflection completed